

FLAKE: Friction and Lattice Analysis of Kinetics and Energetics of rigid 2D clusters on crystalline substrates

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Summary

When a small solid island slides across a crystalline surface, the outcome — smooth flow or stick-slip jerking, free rotation or angular locking — is controlled almost entirely by the geometric relationship between the island and the surface: their relative orientation, size, and lattice symmetry. These phenomena are central to nanoscale friction and to the interpretation of colloidal monolayer experiments (Cao et al., 2019, 2021) and are the premises for the study of *structural superlubricity* in real interfaces (Dienwiebel et al., 2004; Hirano et al., 1991; Wang et al., 2024; Yao et al., 2025; Ying et al., 2025).

FLAKE (Friction and Lattice Analysis of Kinetics and Energetics) is a Python package for simulating the statics and Langevin dynamics of a rigid 2D cluster sliding over a periodic or quasiperiodic substrate potential. It is designed for researchers in nanotribology and 2D-materials physics who need fast, systematic exploration of how cluster geometry, orientation, and substrate symmetry control the energy landscape and friction — studies that are prohibitively expensive with fully atomistic simulations. FLAKE has been the primary simulation tool for three published studies on structural superlubricity and colloidal friction (Cao et al., 2022; Liao et al., 2023; Panizon et al., 2023).

Statement of Need

Nanotribology experiments with colloidal clusters (Cao et al., 2019, 2021, 2022) and van der Waals heterostructures (Liao et al., 2023; Yao et al., 2025) produce rich phenomenology — orientational locking, rotational depinning, coupled roto-translational motion — that is difficult to interpret without a corresponding simulation tool. The standard computational approach in this field is all-atom molecular dynamics with realistic interatomic potentials, typically using LAMMPS (Thompson et al., 2022). While indispensable for elastic effects and chemical specificity, all-atom simulations are prohibitively expensive for the systematic parameter sweeps needed to extract scaling laws and build phase diagrams. They also entangle elastic deformation with the purely geometric commensurability effects that dominate in stiff 2D materials. General-purpose MD packages such as ASE (Larsen et al., 2017) support overdamped integrators but are built around per-atom force fields and do not expose the collective rigid-body geometry needed here.

Recent years have seen a wave of independent computational studies addressing exactly the phenomena FLAKE targets — rotational energy landscapes and scaling laws for twisted bilayers (Zhang & Zhu, 2025b, 2025a, 2026), shape-dependent friction scaling (Gao et al., 2025; Yan et al., 2024), orientational control via strain (Zhou et al., 2025), and experimental evidence for directional superlubricity in van der Waals heterostructures (Le Ster et al., 2024). These studies use a variety of tools and conventions, often relying on LAMMPS with material-specific force fields that constrain the accessible geometries and substrate symmetries. Reviews of

the scaling-up challenge (Ying et al., 2025) highlight that systematic parameter exploration remains a bottleneck. FLAKE provides the missing common ground: a single, geometry- and chemistry-agnostic, Python-native rigid-body package designed for fast, systematic friction studies, and a clean baseline for LAMMPS elastic-contact studies.

State of the Field

Structural superlubricity in 2D materials is an active research front (Wang et al., 2024; Ying et al., 2025). The geometric, rigid-body mechanism is now well-established in principle (Panizon et al., 2023), but quantitative predictions — friction scaling with cluster size and shape, the role of orientation, roto-translational coupling — still require systematic numerical exploration. No existing open Python package targets this combination for the rigid-body tribology use case, leaving each research group to implement its own bespoke simulation protocol.

DFT-level PES tools such as TribChem (Losi et al., 2023) and the high-throughput workflow of Wolloch et al. (2022) operate at the opposite length-scale extreme: they compute interfacial adhesion and shear strength from first-principles electronic structure for periodic unit-cell-sized systems. These tools are naturally complementary to FLAKE — their output (corrugation amplitude ϵ and well shape) can parametrise FLAKE’s substrate potentials for specific material interfaces.

Geometric descriptors such as the Registry Index (Hod, 2013) characterise interfacial commensurability from atomic geometry alone, producing qualitative maps. However, they carry no energy scale and cannot drive dynamics or compute energy barriers.

In LAMMPS simulations of 2D-material interfaces, the interlayer interaction is typically described by the Interlayer Potential (ILP) of Leven et al. (2016), which uses Gaussian-shaped atom-atom overlap terms for registry-dependent repulsion — qualitatively captured by FLAKE’s Gaussian-well substrate, providing a physical link between the two approaches.

Software Design

The central design choice in FLAKE is the rigid-body reduction. A cluster of N particles on a substrate in principle requires $2N$ or $3N$ equations of motion. Treating the cluster as a rigid body collapses this to three collective degrees of freedom — center-of-mass position $\mathbf{r} = (x, y)$ and orientation angle θ — governed by:

$$\eta_t \dot{\mathbf{r}} = \mathbf{F}_{\text{sub}}(\mathbf{r}, \theta) + \mathbf{F}_{\text{ext}} + \boldsymbol{\xi}_t$$

$$\eta_r \dot{\theta} = \tau_{\text{sub}}(\mathbf{r}, \theta) + \tau_{\text{ext}} + \xi_r$$

where $\eta_t = N\eta$ and $\eta_r = \eta \sum_i r_i^2$ are translational and rotational drag coefficients computed from cluster geometry, $\mathbf{F}_{\text{sub}}$ and τ_{sub} are the total substrate force and torque summed over all particles, and $\boldsymbol{\xi}_{t,r}$ are Gaussian white noise terms: $\langle \xi_i(t) \xi_j(t') \rangle = 2k_B T \eta_i \delta_{ij} \delta(t - t')$. The overdamped (inertia-free) limit is the physically appropriate regime for colloidal clusters in viscous media (Cao et al., 2019) and for 2D materials at low sliding speeds; extending the integrator to include inertia is straightforward. This reduction makes it possible to run thousands of trajectories in minutes on a laptop, enabling the systematic parameter sweeps — over cluster size, shape, orientation, substrate symmetry, and applied force — that are the primary target of FLAKE.

The rigidity assumption is justified by the physics: for stiff 2D materials (graphene, hBN, MoS₂) at moderate loads, and for colloidal clusters where interparticle distances are effectively fixed, the dominant friction mechanism is geometric commensurability, not elastic deformation

(Panizon et al., 2023). **Limitations.** The rigid-body approximation does not capture intralayer elasticity, which drives the Aubry-type commensurate–incommensurate transition (Wang & Tosatti, 2025); moiré-induced out-of-plane buckling in freestanding bilayers (Wang & Tosatti, 2024); or elastic edge effects that give rise to chiral moiré dynamics at finite contacts (Silva et al., 2024). These effects require fully elastic simulations and are explicitly beyond FLAKE’s scope.

The performance design follows directly from this reduction. All substrate energy and force kernels are Numba just-in-time compiled (@njit, Numba’s JIT decorator (Lam et al., 2015)): the `substrate_from_params` factory pre-converts parameters to float64 arrays once and returns closures, eliminating per-step Python overhead, and `run_md` dispatches to a fully JIT-compiled inner loop when no Python callbacks are needed (10–50× faster than an interpreted loop for long trajectories).

The string method for minimum-energy paths (E et al., 2002) is extended to the joint (x, y, θ) space using independent length scales (l_x, l_y, l_θ) that put translational and angular displacements on equal footing — a non-trivial metric choice required to find physically meaningful roto-translational paths.

Features. FLAKE provides:

- *Energy landscape maps* over translational and/or rotational degrees of freedom (`flake map`), for static analysis of commensurability and orientational locking.
- *Minimum-energy path search* via the string method in 2D (translational) and 3D (roto-translational) configuration space (`flake string`), giving static friction forces and transition pathways.
- *Parameter-sweep Langevin dynamics* (`flake sweep`) parallelised over a parameter grid via `joblib`, for depinning thresholds and velocity–force scaling laws.
- *Substrate types*: sinusoidal (optical lattices), Gaussian well (ILP-like registry dependence), tanh well (lithographic patterns), and flat (testing baseline); new types require only a single Numba @njit function.
- *Cluster geometries*: circle, hexagon, triangle, rectangle, ellipse, and arbitrary polygon via `Shapely`; multi-site bases supported.
- A *YAML-driven CLI* and six end-to-end Jupyter notebooks in `examples/` covering the full workflow from substrate construction to scaling-law analysis, with no Python scripting required.

Research Impact

FLAKE v0.1.3 is the formalised, tested, and packaged release of a codebase developed from 2022 onward at SISSA. The predecessor (v0.0.6, publicly available since November 2022) was the primary simulation tool for the following published studies:

- Panizon et al. (2023) (Nanoscale): geometric classification of arbitrary 2D interfaces and scaling laws of static friction force from finite commensurate and incommensurate contacts.
- Cao et al. (2022) (Physical Review X): coupled roto-translational depinning and the role of moiré pattern evolution in colloidal experiments.
- Liao et al. (2023) (ACS Applied Materials and Interfaces): combined experiment and simulation of twisting dynamics in large lattice-mismatch van der Waals heterostructures.

The colloidal experiments of Cao et al. (2019) and Cao et al. (2021) provided the physics foundation and motivated the initial design of the package.

Version 0.1.3 adds a fully JIT-compiled MD inner loop, the roto-translational string method, and a complete test suite. The package is currently in active use at SISSA for ongoing research on quasicrystalline substrates and roto-translational synchronisation effects, in collaboration with experimentalists in Germany and China.

AI Usage Disclosure

The physical model, all design decisions, and the research direction of FLAKE are entirely the work of the author. The central argument — that geometry dominates friction in stiff 2D contacts, making a rigid-body model both sufficient and necessary for systematic study — is grounded in the author's prior publications (Cao et al., 2022; Panizon et al., 2023). The choice of substrate models, the roto-translational string method and its arc-length metric, and the scope of the package all predate any AI involvement: the core physical model and simulation algorithms were functional in the predecessor codebase (v0.0.6, November 2022) prior to any AI assistance. Claude Code (Anthropic, claude-sonnet-4-6) was subsequently used to improve the codebase: making the code more readable (docstring rewriting, module restructuring and API cleanup), more efficient (Numba JIT refactoring of substrate closures and the MD inner loop), and adherent to modern open-source standards (test suite scaffolding and CI configuration) and improving this manuscript. All AI-generated outputs were reviewed and edited by the author.

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